

High-quality and high-volume test data generation through End-to-End simulators for Earth Observation: A CHIME case study

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Abstract

Next-generation Earth observation missions, such as the Copernicus Hyperspectral Imaging Mission for the Environment (CHIME), face a critical bottleneck: the generation of high-volume, high-quality Test Data Sets (TDS) within reasonable timeframes. End-to-End (E2E) simulators are the primary tools used to generate such data. Achieving the required quality demands a high level of physical and modelling detail within these simulators. This detail is formalized in Algorithm Theoretical Basis Documents (ATBDs). Experience shows that during the design and detailed implementation of simulators, a translation gap often exists between the mathematical definitions captured in the ATBDs and their software realization. This gap makes it difficult for system and instrument performance engineers to assess the correctness of the implementation or to modify modelling assumptions without triggering lengthy software development cycles. ESA provides comprehensive high-level guidance for E2E simulator development through Reference Architecture, Interface, and Generic Requirements documents, alongside frameworks like Biblos3. However, this high-level focus often leads to rigid, mission-specific implementations that limit reuse and obscure the underlying physics from systems and instrument engineers. This paper reviews current E2E simulation practices in the ESA Earth observation context, identifying areas where traditional development could be improved for faster data generation, increased flexibility, maintainability and validation. Key functional and non-functional requirements are identified, together with common pain points affecting reproducibility and testability. A comparison is drawn with alternative E2E simulation approaches used in large-scale scientific software systems, such as the ATLAS Athena framework at CERN. As a major contribution, a CHIME Instrument Optical Module (IOM) prototype demonstrates a functional, Python-centric approach, aligning implementation closely with ATBDs. Building on these lessons, we propose a standardized two-pillar architecture that natively supports GPU acceleration. These pragmatic recommendations aim to simplify development, leverage modern hardware for faster TDS generation, and reduce long-term maintenance, verification, and validation efforts.

1 Introduction

End-to-end (E2E) mission performance simulators (E2ES) are indispensable tools in the development of modern Earth observation (EO) missions at the European Space Agency (ESA) [1, 2, 3, 4, 5], and are increasingly being adopted for projects at other agencies, including DLR and JAXA [6, 7]. These simulators are critical for assessing mission performance and verifying ground processing chains [2, 3, 4]. To ensure interoperability, ESA has successfully standardized the high-level architecture of these systems, particularly regarding module interfaces and the overarching framework provided by openSF [8, 9, 5, 10].

However, while external interfaces are well-defined by ESA, the internal implementation of these modules remains highly flexible, with lack of formalized guidance. Aspects such as coding style, library selection, and internal software architecture are often left to the discretion of individual contractors, although some specific recommendations exist, such as the use of OpenMP and OpenCL [5, 11]. This lack of standardization frequently leads to monolithic codebases where standard operations—such as convolutions, downsampling, or binning—are implemented directly within the mission-specific logic. Consequently, extracting and reusing code for subsequent missions can become challenging, as these standard functions become tightly coupled with the project-specific architecture.

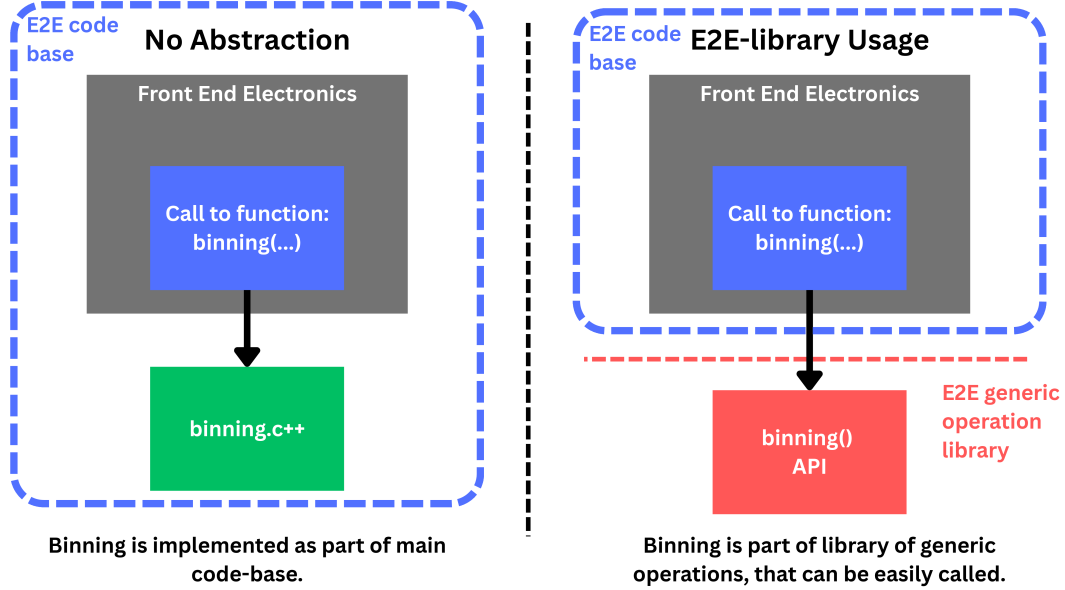


Figure 1: Comparison of implementation strategies. The traditional "No Abstraction" approach (left) embeds standard operations directly into the mission-specific codebase, requiring repeated testing and adaptation. The proposed "E2E-library Usage" approach (right) delegates these operations to a validated, external API, promoting reusability and keeping the main codebase lean.

These architectural challenges have practical consequences beyond software maintenance. The high complexity and limited transparency in the code create significant challenges for Instrument and System Performance engineers. These specialists, who often have backgrounds in physics and engineering rather than software engineering, are responsible for validating the underlying scientific algorithms [2]. When implementation details are obscured by overly complex software architecture, the gap between the physical model and its digital twin widens, increasing the risk of errors and driving up development costs.

The goal of this paper is to review current implementation and technological practices in the ESA EO context, identify core requirements and limitations, and propose simplifications aimed at reducing development and testing effort. Building upon existing ESA guidelines and the Biblos framework [11, 5, 12], this work incorporates insights from a recent ESA white paper and internal discussions. We compare current practices with alternative large-scale scientific frameworks, such as ATLAS Athena at CERN, to identify transferable lessons.

Finally, we propose a Python-centric architectural direction designed to produce cleaner, more readable code that helps to serve as its own documentation and supports multi-hardware options, ultimately facilitating easier validation for ESA staff and more efficient delivery for industrial contractors.

2 Background and Scope

To establish a common baseline, it is necessary to define the functional boundaries of the systems under discussion. According to the ESA E2E White Paper [2], an End-to-End Mission Performance Simulator (E2ES) is defined as:

"...chains of software tools developed to simulate the generation of the geophysical observables target scene, the whole observation process from space including orbit, pointing and guidance, the instruments, the on-board data generation and the processing of the outputs of these instruments to retrieve the estimate of the original observables."

In this context, the E2ES serves as a "digital twin" of the entire measurement chain. While the high-level orchestration of these chains is often handled by frameworks like openSF, this paper focuses specifically on the **module-level architecture**—the internal design and implementation of the individual building blocks (e.g., Instrument Simulation Modules) that constitute the chain.

2.1 Scope of the Discussion

The scope of this work is centered on the software engineering and operational aspects of E2ES development rather than the underlying physics. Specifically, we address:

- **Software Architecture and Technology:** Reviewing the current use of object-oriented module architecture, languages, and libraries, with a focus on moving toward a Python-centric stack that utilizes modern scientific computing tools developed in performant languages
- **Hardware Utilization:** Discussing the trade-offs between CPU-based processing and GPU-accelerated workflows to meet increasing performance demands.
- **Validation and Testing Workflows:** Proposing methods for automated error reporting and result analysis to replace labor-intensive manual verification and to properly analyze numerical errors.

- **Documentation Practices:** Exploring the transition from static, text-heavy implementation documentation toward self-documenting code and markdown documentation as part of the code repository.

Conversely, this paper does not cover mission-specific algorithm derivations or detailed physical modeling for particular Earth observation sensors.

Finally, the paper references a practical implementation case: the Instrument Optical Module for the CHIME (Copernicus Hyperspectral Imaging Mission for the Environment) E2ES. While this module did not adhere to all the proposed guidelines from its inception, the lessons learned during its development served as the catalyst for the requirements and architectural directions proposed in the following sections.

2.2 Target Audience

The recommendations provided here are intended for software implementation contractors, scientific algorithm developers, instrument performance engineers, simulator and processor engineers, and ground segment system engineers. The objective is to suggest modern tools and workflows for industry partners, while outlining expected technical standards for ESA staff. By lowering the implementation complexity, we aim to provide ESA staff with more transparent tools for verification while reducing the development burden on industrial partners.

3 Current E2E Module Architecture and Implementation

3.1 High-level overview of architecture

The highest-level architecture of E2ES within ESA can be described primarily by the functional role of its constituent modules:

- **Instrument Performance Module** - responsible for simulating satellite and instrument behaviour.
- **Ground Prototype Processor** - L1 processing to L2 retrieval.
- **Performance Assessment Module** - responsible for comparing the input to the retrieved image in order to assess mission performance requirements.

This high-level structure is further decomposed into several modules, an example of this dataflow is illustrated in Figure 2.

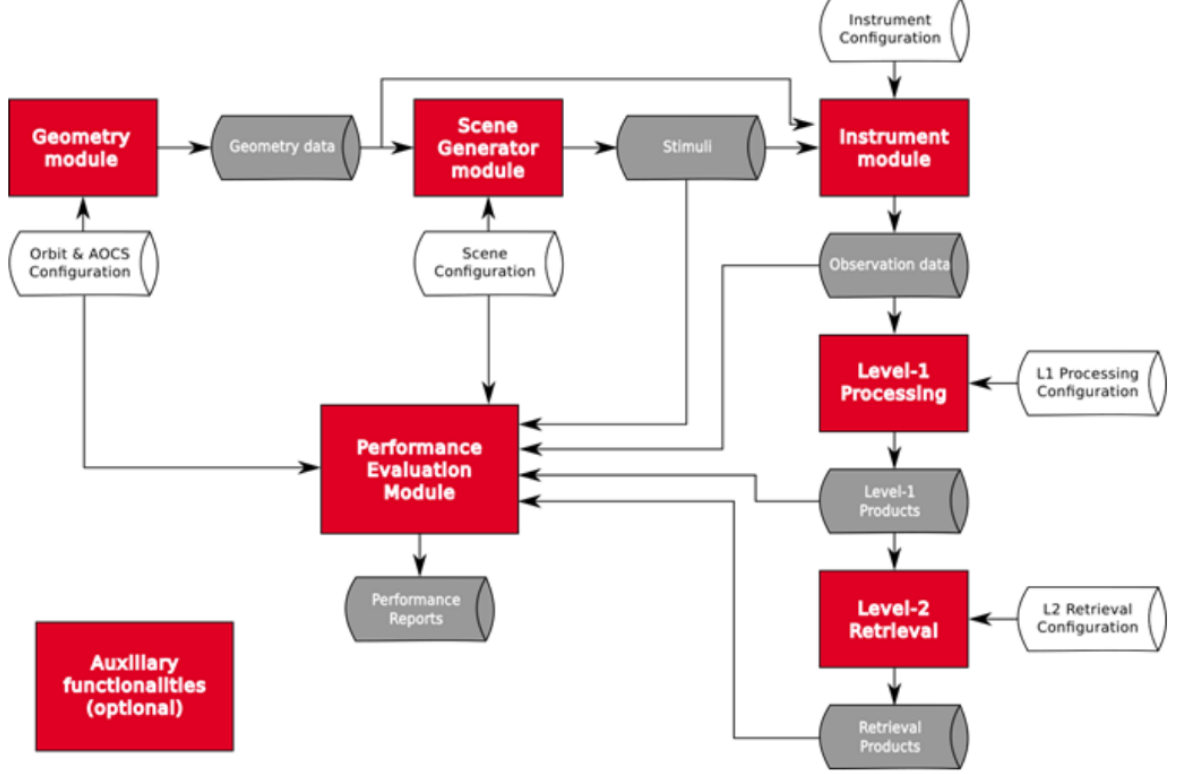


Figure 2: Generic dataflow at highest level of architecture [10].

While generic implementations of these modules exist for academia thanks to the Biblos3 project [5], their specific implementations must always be tailored to meet mission-specific requirements. It is at this implementation level that this paper focuses its attention.

3.2 Common Development Phases

The development of E2ES software follows an evolutionary approach that spans the entire mission lifecycle, progressing from simplified feasibility models to detailed, mission-specific implementations [2]. This progression can be characterized by several distinct stages:

1. **Initial Generic Implementation (Phase 0/A/B1):** A generic simulator is developed to assess mission feasibility and L2 retrieval concepts. This early version typically employs simplified error models applied to Scene Generation outputs [2].
2. **Mission-Specific Development (Phase B2/C/D/E1):** Detailed, mission-specific implementations are developed. While these may occasionally build upon the early

generic foundation, they are frequently developed independently under separate contracts, often by entirely different industrial consortia. Separate ATBDs are developed for individual modules (ISM, L1 processing, L2 retrieval), and the software is then implemented according to these algorithmic specifications [2].

3. **Validation and Test Data Generation:** Throughout development, Test Data Sets (TDS) are created for systematic verification and validation of both individual modules and the integrated chain [2].
4. **On-Ground Calibration Support:** The E2ES supports On-Ground Calibration Activities, where measured data is used for characterizing instrument behavior and preparing final IOM module [2].
5. **In-Orbit Calibration and Verification:** During the commissioning phase, the GPP component of the E2ES is utilized to support In-Orbit Calibration and Verification (IOCV) activities, ensuring the system operates correctly post-launch.
6. **Operational Ground Processing (Phase E2):** The toolchain transitions to supporting Ground Processor maintenance, algorithm evolution, and anomaly investigation during long-term operations [2].

If not managed properly, this phased approach may easily result in a software architecture that leaves concepts such as maintainability and testability as secondary priorities. As the codebase evolves across different mission phases and industrial teams, the natural accumulation of project-specific adaptations could sometimes create tension between rapid implementation and long-term development, directly motivating the architectural concepts proposed in this paper.

Furthermore, it is critical to recognize the computational nature of E2ES software. In most cases, these are image processing applications that handle very large datasets. Given that processing images of this scale is both memory-intensive and computationally demanding, these modules are typically implemented in highly performant C/C++ [5, 10, 11], with parallelization support provided through OpenMP and OpenCL [11, 5].

3.3 Requirements

Functional and non-functional requirements for E2ES software are specified by ESA in [11]. Key requirements include:

- E2E-ARC-GEN-080 - Code shall be parallelized
- E2E-ENV-GEN-060 - Implementation in C or C++14 or newer

...it is explicitly required that in Phase B2/C/D the SW is implemented in a performant compiled language.

- E2E-ENV-GEN-090 - Ground processing parallelization through the use of OpenMP
- E2E-ENV-GEN-110 - GPU support through common APIs (e.g., OpenCL, CUDA)

3.4 Recurring Limitations and Pain Points

While the requirements outlined in Section 3.3 are understandable from a performance perspective, they often impose limitations that are not always aligned with practical development realities. Performance is not the only consideration—development cost, ease of maintenance, and accessibility to domain experts are equally important. This clash of domains is particularly evident in environments where ATBD preparation, software development, and validation are distributed across different locations and split between ESA and industrial partners [2].

Several recurring challenges have been identified through recent mission developments and internal discussions, notably within the CHIME and LSTM teams, who have advocated for more flexible technological approaches, especially in the context of Instrument Simulator:

1. **Language and Ecosystem Constraints:** Python is currently the most popular programming language, with extensive support for scientific computing and a rich ecosystem of data science tools [13, 14]. While ground processing chains may require peak performance, achieving comparable performance for Instrument Simulation Modules (ISM) or Scene Generation Modules (SGM) using Python with modern libraries has the potential to significantly reduce development costs. This is especially relevant given that many system experts routinely use Python but lack deep C++ expertise.
2. **GPU Acceleration:** Although requirement E2E-ENV-GEN-110 mentions GPU support, there are cases where the adopted architecture makes no use of the potential presented by GPU processing, leaning towards more conventional architectures that rely heavily on data transfer, thereby favoring CPUs. While industry conclusions in early project phases (e.g., five years ago) may have validly dismissed GPUs due to the technological landscape and integration complexities of the time, the rapid advancement of GPU hardware and scientific computing ecosystems makes revisiting these trade-offs a priority today. A stronger challenge must be presented in the early phases of new projects to ensure this potential for orders-of-magnitude performance improvement is not lost from the start.

This potential is highly evident when benchmarking core simulator operations. To demonstrate this, preliminary experimental tests were conducted, evaluating fundamental E2E operations such as convolutions, gain application, and electronic offset

corrections. As detailed in **Appendix A**, these tests demonstrate an average processing time reduction of **X orders of magnitude and an X -fold** improvement in cost-to-performance ratio when comparing modern GPU processing (e.g., NVIDIA H100) to standard CPU processing (e.g., 16-core Intel Xeon). While GPU usage can be limited by PCIe memory bandwidth [15], most use cases can mitigate this by pipelining multiple operations entirely on the device, avoiding excessive host-device data transfer [16, 15]. Providing both CPU and GPU execution paths—even if the GPU path only covers partial, computation-heavy pipelines—would offer critical flexibility, particularly when intermediate breakpoint files are not required during nominal runs.

Comment

Check the appendings, numbers to be filled.

3. **Performance Monitoring:** Automated support for timeliness metrics and runtime performance visualization could be included as part of the baseline.
4. **Manual breakpoint validation and numerical error analysis:** The absence of automated report generation for results analysis and numerical error propagation at breakpoints places an unnecessary burden on validation teams. Automated tooling, implemented at the scientific library level, for these tasks would reduce manual effort and improve reproducibility.

Comment

I think I understand this better now.

So you have two things.

- One that operates at "breakpoint" level, that allows to easily access the input vs output of some operation.
- Another one that operates at function level, that allows to assess the numerical error introduced in the mathematical operations due to the selection of data types.

Am I right?

If so, I would change the name of this point to:

" Manual breakpoint validation and numerical error analysis"

Answer

Yes exactly, with slight difference, the numerical error is assessed at breakpoints. The algorithmic library will have something similar but at function level as you said.

5. **Hardware-Software Co-Design:** While the OpenSF framework proposes the use of a file-based interface between modules, there are cases where it can be convenient to deviate from it since significant performance improvements could be obtained by making use of fast memory interfaces, optimized I/O strategies, and detached save threads. Similar to the need for stronger early focus on architectural decisions mentioned regarding GPUs, these hardware implications must be evaluated from the project's origin.
6. **Complexity Barriers for Domain Experts:** High levels of implementation complexity at the ISM level can directly impede the productivity of Instrument Performance engineers and system experts, who are responsible for validating the underlying physics. A key architectural challenge early in projects is the widespread reliance on heavily nested Object-Oriented Programming (OOP) paradigms, which can sometimes abstract algorithmic logic behind complex state management. Shifting towards a functional programming approach for physically meaningful operations—utilizing pure functions with explicit inputs and outputs rather than mutating references—improves code readability for non-software experts. This approach enables experts to work more efficiently and significantly simplifies the direct translation of mathematical ATBD equations into software implementations, without sacrificing modeling accuracy.
7. **Documentation-Implementation Gap:** Traditional documentation is prone to diverge from the actual implementation, introducing validation risks. Adopting self-documenting code practices—where clear structure and meaningful naming serve as the primary reference—mitigates this discrepancy. Highly readable code would enable these models to automatically parse and cross-check the programmatic implementation directly against mathematical ATBD specifications.

Properly addressing each one of these challenges can help to reduce development time, increase code transparency and ease the validation effort. A shift towards pragmatic architectural choices that balance performance, usability, and long-term maintainability is strongly advocated in this paper.

4 Alternative E2E Approaches: Lessons from Athena

This section presents a high-level comparison with E2E simulation practices adopted in large-scale scientific software systems, with particular focus on the ATLAS Athena framework at

CERN. Key aspects such as Python-based configuration, collaborative development models, specialized tooling, and documentation practices are discussed. Relevant lessons and transferable practices for ESA Earth observation simulators are highlighted.

4.1 Python-Based High-Level Configuration

CERN acknowledges the need for Python tooling for high-level data analysis [14, 17, 18]. In Athena, high-level configuration—equivalent to defining specific processing pipelines such as the IOM in ESA context—is performed at the Python level [17, 18]. This approach allows scientists and engineers to assemble and customize components without requiring C++ algorithm modifications. Critically, these components can be reused across different experiments and analyses, reducing duplication of effort. This configuration strategy directly addresses the accessibility and flexibility needs identified for ESA E2ES implementations.

4.2 Open Collaborative Development Model

A distinguishing feature of Athena is its development model: the framework is collaboratively developed by CERN staff and academic researchers worldwide, fostering extensive peer review. In contrast, due to necessary contractual frameworks, at ESA, the involvement of the scientific community and software developers is typically restricted to those within the specific contracted consortium.

Adrians comment

I am still not convinced, to review this sentence

Testing, development, and integration in Athena are conducted through GitLab in an open-source manner. While Git-centric approaches are already standard within individual ESA projects, ESA personnel primarily operate as reviewers rather than active coders. Therefore, the most beneficial advancement would be establishing a centralized, cross-ESA Git repository hosting all E2E simulators. This would enable ESA and future consortia to easily extract and reuse validated code, significantly improving knowledge transfer and reducing redundant efforts across missions.

Adrians comment

I am still not convinced, to review this sentence

Answer

I will leave it as it is now as I'm not sure to what to change it to.

4.3 Specialized Libraries and Tooling

CERN relies heavily on ROOT, a specialized C++ library for data management, and Gaudi, an event-processing framework. For ESA applications, developing an analogous, mission-agnostic algorithmic library could yield significant benefits. Built upon established tools like OpenCV or Dask, and functioning as a high-performance extension of NumPy, this library could provide:

- Built-in automated report generation
- Automated numerical error testing
- Integrated logger, timer and other utilities
- Clean, consistent interfaces for algorithm implementation

Supporting both direct C++ implementation and Python access via bindings (e.g., pybind11) would offer critical flexibility while maintaining a unified development experience. Furthermore, this could leverage parallel computing libraries such as Dask, harnessing their computational power while abstracting the associated development complexity.

4.4 Documentation Practices

Athena maintains comprehensive, web-based documentation (accessible at <https://atlas-software.docs.cern.ch/>). Adopting a similar GitLab-based Markdown documentation system for ESA E2ES would be highly beneficial, particularly for scientific users and validation teams requiring rapid guidance on setup and workflows. This approach consolidates scattered information and ensures stakeholders always access the most current version. Furthermore, version-controlled Markdown makes correcting documentation errors significantly more efficient than drafting and distributing traditional PDF revisions.

5 Python-Centric Prototype: Lessons from CHIME IOM

As part of a recent internship with the CHIME team, a prototype Instrument Optical Module (IOM) was developed using a Python-based stack centered on Dask, Xarray, NumPy, and Numba [19, 20, 21, 22]. The objective was to provide a more flexible alternative to existing C++ implementations, particularly for supporting the configuration of the instrument Optical Chain with data coming from On-Ground Calibration Activities. To emphasize readability and maintainability, the architecture deliberately departed from traditional Object-Oriented Programming (OOP) paradigms in favor of a strict functional programming approach across all implementation levels. A generic form of this architecture is shown in Figure 3.

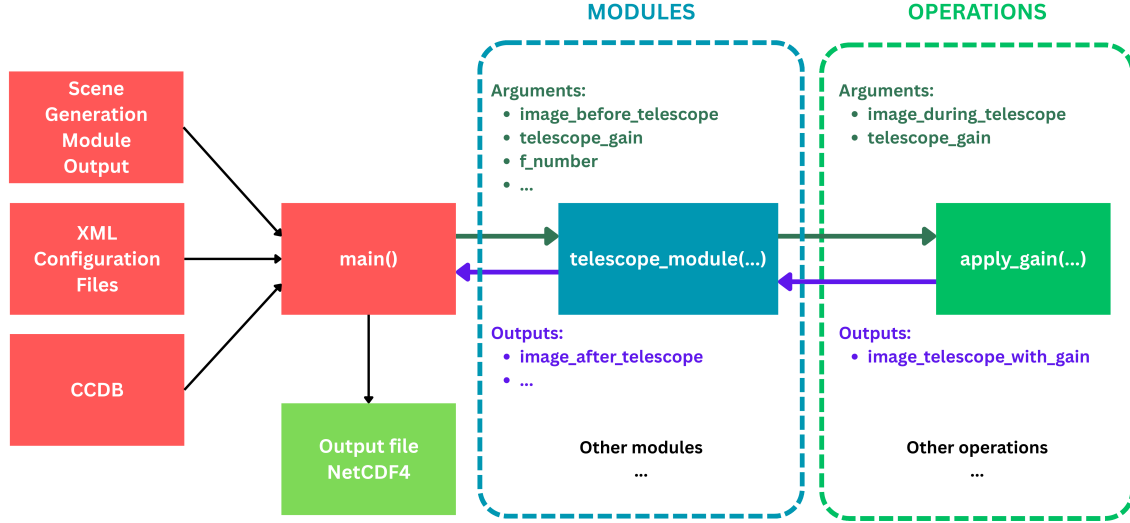


Figure 3: Simplified generic dataflow of the Instrument Module; modules are written primarily as pure functions, providing clear signatures for both the input and output.

5.1 Observed Strengths

The prototype demonstrated several advantages:

1. **Memory Efficiency Through Chunking:** Dask’s native chunking support enabled significant reductions in RAM requirements [20]. Initial input images exceeding 50 GB—which could drive total memory consumption beyond 128 GB during processing—were successfully handled on virtual machines with only 32 GB of RAM. This capability would enable simulator deployment on standard computing infrastructure rather than requiring specialized high-memory systems.

	Array	Chunk
Bytes	5.86 GiB	600.00 MiB
Shape	(512, 3072, 1000)	(512, 3072, 100)
Dask graph	10 chunks in 2 graph layers	
Data type	float32 numpy.ndarray	

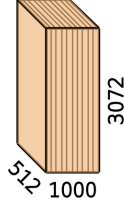


Figure 4: CHIME unbinned example of Dask chunking (SPE, ACT, ALT) [23]; the image is processed as slices of a data cube, for example sliced in spectral bands, which are stored as standard NumPy arrays.

Adrians comment

You can show the dimensions (ACT, ALT, SPE) and in which dimension the chunking was done (in our case, it was the spectral dimension).

Answerk

I updated, but generally chunking was done not for whole run, I think starting from Spectrometer(after Slit block) we worked with ALT chunking.

2. **Algorithm Clarity:** Implementation using NumPy and scientific Python libraries resulted in substantially shorter and more transparent code compared to equivalent C++ implementations.
3. **Simplified Data Management:** Dask’s lazy evaluation model streamlined checkpoint (BKP) file generation, consolidating multi-step save operations into single function calls.
4. **Comparable Performance:** Preliminary benchmarks on a representative test case showed processing times of 60–105 minutes for the Python implementation versus approximately 75 minutes for the nominal C++ version, suggesting performance parity for certain workloads. However, more comprehensive testing across diverse scenarios is required for validation.
5. **Extensibility:** Performance-critical functions can be selectively reimplemented in C++ or CUDA and integrated via Pybind11 [24], enabling targeted optimization while preserving Python’s high-level expressiveness. However, non-standard functions can also be implemented using Numba, which provides highly efficient JIT-compiled code for both CPU and GPU execution [22], while retaining Python’s simplicity. In cases

where C++ implementation was utilized, pure Python fallback implementations were also provided, with the structure shown in Figure 5.

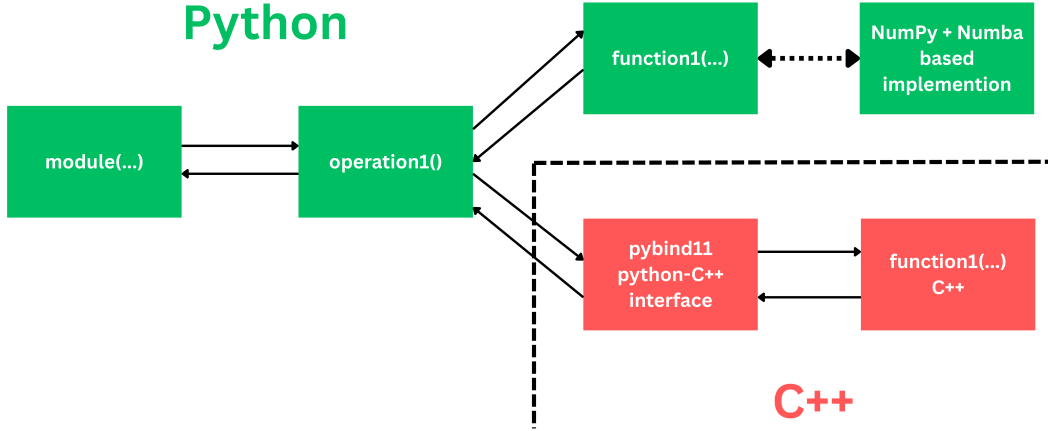


Figure 5: Function interfaces for the Python-C++ implementation.

Supporting On-Ground Calibration (OGC) activities is typically handled by integrating specific execution paths directly into the main processing pipeline. However, to avoid modifying the core codebase while detailed OGC requirements were still undefined, an alternative wrapper script approach was tested in the prototype.

As illustrated in Figure 6, this script utilized "monkey-patching" to dynamically alter code at runtime. By acting as an external runner, it modified how the simulator perceived the Characterization and Calibration Data Base (CCDB), overrode specific function signatures, and injected new parameters on the fly, providing extensive configuration freedom without altering the underlying `pysim` codebase.

While this method successfully isolated OGC modifications, the resulting dynamic complexity was significant. Upon reflection, the standard practice of embedding OGC-specific logic inside the main pipeline is generally simpler and more maintainable. Although this requires modifying the core code, the added internal complexity could likely be easily managed if the top-level architecture utilizes the explicit, functional approach discussed earlier.

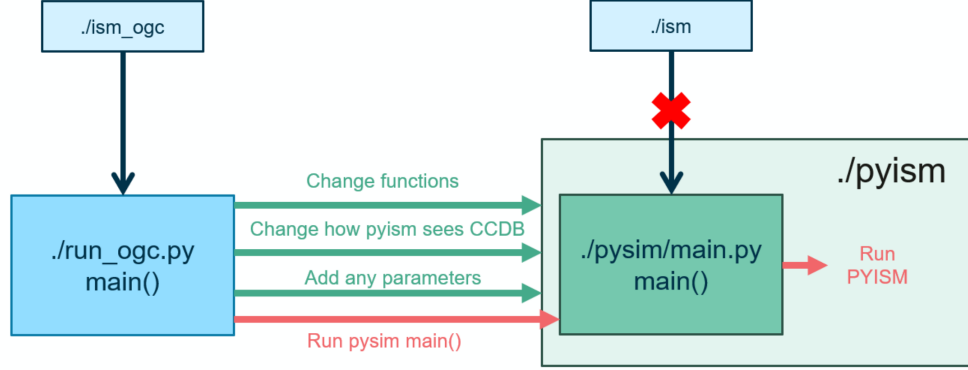


Figure 6: Alternative OGC support architecture using an external wrapper script. The `run_ogc.py` script dynamically patches functions, modifies CCDB visibility, and injects parameters before execution, avoiding direct modifications to the core `pysim` codebase.

5.2 Identified Limitations

Several challenges emerged during development:

1. **Error Diagnostics:** Dask generates verbose stack traces that can obscure the root cause of failures, complicating debugging efforts.
2. **Backend Instabilities:** The netCDF4 and HDF5 backends occasionally failed when parallel workers attempted concurrent writes, requiring workarounds with zarr.
3. **Overhead and Control:** Dask’s multiprocessing layer, coupled with Dask graph generation, introduces significant data size-independent overhead and limits low-level execution control compared to native parallelization frameworks like OpenMP.

Overall, this experiment suggests that a dedicated, high-performance library, written in C++/CUDA and equipped with essential utilities, could significantly accelerate simulator development while providing the necessary infrastructure support.

A similar library, featuring a C++ backend API tailored specifically for the ESA E2E use case, could also be employed. In this context, chunking may not be strictly necessary, as images handled during ground processing are typically of significantly smaller size than those ingested by the IOM.

6 Discussions and Recommendations

The analyses in the preceding sections acknowledge that current E2E simulators are highly complex software systems that have been reliably delivered using established approaches.

However, the Python-based implementations explored in this study present a viable alternative capable of addressing underlying systemic issues related to maintainability, transparency, and testability.

As a major contribution of this work, the CHIME IOM prototype establishes a proof of concept for Python-centric ISM E2ES architectures. While it successfully demonstrates the benefits of a functional, Python-driven approach, further work is needed to develop robust, production-ready frameworks. To address these systemic issues at an agency-wide scale and shift away from isolated, project-specific implementations, we propose the adoption of a standardized, two-pillar library architecture.

6.1 A Two-Pillar Library Architecture

Rather than building simulators based on generic modules and frameworks for each mission, future developments should rely on two distinct, complementary libraries to strictly separate mathematical implementation from pipeline orchestration.

6.1.1 The E2E Algorithmic Library

The first proposed pillar is a purely algorithmic library that abstracts standard Earth observation processing operations (e.g., convolutions, resampling, gain applications or ADC conversion). This library would provide standardized endpoints for both Python and C++, with seamless underlying support for CUDA to enable GPU acceleration for performance-critical operations.

Adrian Comment

to review this proposal

This algorithmic library directly synergizes with the proposed cross-mission ESA GitLab repository discussed previously. Consortia and ESA engineers could extract well-tested monolithic code from past E2E simulators, write generic wrappers, and contribute them to this centralized library. As introduced in Figure 1, decoupling these operations into an abstract library provides concrete benefits:

- **Reduced codebase size:** Mission-specific repositories remain lean, focusing purely on high-level orchestration rather than low-level mathematical implementations.
- **Elimination of redundant testing:** Once a generic function is validated and approved on the ESA side, it can be safely reused across projects without requiring repeated unit testing at the project level.

- **Cross-mission reusability:** Industrial partners can easily port fundamental functionalities to new mission requirements, significantly reducing development time and costs.

The level at which these interfaces are proposed is demonstrated in Figure 7. The library would provide simplified interfaces for array handling without attempting to rewrite established tools like Xarray or Dask, while offering standardized testing interfaces to easily visualize data.

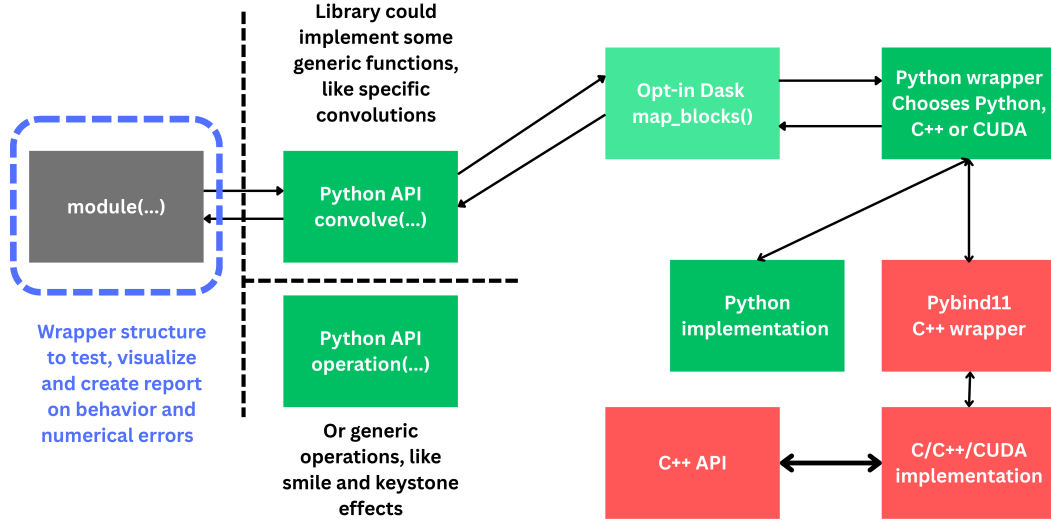


Figure 7: Example architecture of the proposed Python/C++ E2E Algorithmic Library. It provides simplified interfaces for common functions, automated testing, and visualization, abstracting the implementation complexity away.

6.1.2 The E2E Framework Library

The second pillar is a framework library dedicated strictly to handling E2E-specific software infrastructure. Instead of each consortium providing their own implementations of parsers and I/O handlers, this framework would manage standard APIs, ingest required configuration files (e.g., OpenSF compliance), and provide an intuitive interface to assemble new pipelines using the abstracted components from the Algorithmic Library.

Crucially, this framework would natively support automated numerical error propagation throughout the entire processing chain. It would standardize simulator outputs by automatically tracking timeliness metrics, performing automated Quality Assurance for numerical stability, and generating customizable, python script reports, capable of generating automated

PDFs, at each pipeline stage. This would directly address the limitations in transparency and performance monitoring identified in current operational implementations.

6.2 Summary of Recommendations

Adrians Comment

You are missing - a recommendation to depart from an Object Oriented Paradigm in favour of a Functional Paradigm - a recommendation to go to self documenting code and markdown repositories, mainly for DPMs (DPM stems from Detail Processing Model and are documents that specify how an ATBD has been implemented in code)

Answer

Added.

Based on the preceding analysis, the following specific actions are recommended for future E2E developments:

1. Adopt a Python-centric hybrid stack by utilizing the proposed two-pillar library architecture, using Python for configuration and pipeline assembly while invoking compiled C++/CUDA algorithms for heavy image processing.
2. Prioritize a functional programming paradigm over heavily nested object-oriented architectures for core algorithmic modules.
3. Enforce hardware-software co-design from early project phases, explicitly evaluating GPU acceleration and optimized I/O strategies to prevent performance bottlenecks.
4. Standardize simulator telemetry by mandating automated timeliness tracking, numerical error propagation, and automated QA report generation natively through the framework library.
5. Establish a centralized, shared ESA Git repository to extract common algorithms, facilitate continuous review, and build the foundation for the cross-mission Algorithmic Library.
6. Adopt self-documenting code practices and Markdown-based Detailed Processing Models (DPMs) integrated directly into the code repository, replacing static external documents
7. Leverage AI validation tools to verify consistency between the readable, functional codebase and the mathematical ATBD specifications.

7 Internal Physical Modeling Suggestions(FOR CHIME TEAM ONLY - Will be removed from paper)

The complexity of the IOM implementation does not really come from the underlying physics, which are relatively straightforward, especially when ATBD is already provided. Aside from specific non-linear detector effects, ATBD primarily relies on four standard mathematical operations:

- 1D and 2D resampling (interpolation).
- 1D convolution over 2D arrays.
- 2D convolution over 1D arrays.
- Element-wise multiplication.

Despite mathematical simplicity of most operations, the current software implementation is often difficult to verify. This difficulty arises from the low-level management of grids and margins. Almost every operation requires re-calculating indices or handling margins manually, which leads to inconsistent implementations and potential numerical errors. Good example is usage of `arange` function for `float32`, which in our case generated the numerical error of 0.006% for final elements, compared to linear space (`np.linspace`). This error was only error seen at `PyISM` and `OPSI` functions correctly.

To address this, I would propose two architectural simplifications, tho it is likely not feasible to refactor the existing code now:

- **Standardized Grid Abstraction:**
The handling of array margins should be abstracted into a single, standardized function. Currently, different parts of the code calculate margins differently(`telescope LSF` has margins of 2, while `HSI LSFs` have only of 1). From a physical perspective, minor differences in margin inclusion should not significantly impact the result(as we dont know what happens there as it was not measured); if they do, it implies an error in how boundary is modeled. Standardizing this logic removes the risk of "off-by-one" errors and index mismatches.
- **Early Resampling Strategy:**
The current architecture performs spectral resampling from SGM to oversampled detector grid late in the chain (at the Detector module level), requiring more complex convolution logic to align data(this perhaps is one of the hardest parts of code to understand). I suggest moving this resampling step to the very beginning of the pipeline,

immediately after the Scene Generation Module (SGM) in Motion module. By ensuring the input data matches the final target spectral grid from the start, we can eliminate the complex, error-prone code currently required to handle coordinate transformations later in the chain.

To conclude, the hard part of the simulator, is not the physical modeling(except Smile and keystone which are most complex operations at IOM level) when you have access to ATBD, but often the implementation of coordinate transformations. By abstracting grid management and optimizing the order of operations, the code could becomes more readable and less prone to the hidden indexing and numerical errors.

References

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